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Chapter 6

Production Performance Reports



Machine Efficiency



This report provides an analysis of how well your production resources are performing against the production standard on the planning route. The report details can help you to identify those machines that are failing to meet the standard.

In addition, the report provides information regarding the costs of non-conformance arising from efficiency variances.

You can:

- Produce the report for any time period, providing machine information has been retained for the date range selected.
- Restrict information to a single shift in the period.
- Include or exclude rework from the report.
- Analyse machine efficiency in terms of labour or machine performance.



Select Machine Efficiency on the menu to display the Work Station Efficiency Report Selection screen.

MJ600	ZS - Style Demonstration Co.	User GBJBMS	29/03/96 11:51:54
Work Station Efficiency Report Selection			
Organisational Model...? __			
WIP Locations From....? __ to __			
Work Stations From....? ____ to ____			
Shift Number..... __			
Date range from..... 29/03/96 to 29/03/96			
Report Basis..... 2 1=Machine, 2=Labour			
Include Rework..... 1 1=Yes			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35 SF MW KS IM II KENB20A PET166S2			

Fields:

Organisational Model

Enter the 'live' organisational model.

WIP Locations Range

Enter a range of WIP locations or leave blank to include all locations.

Machine Range

Enter a range of machines or leave blank to include all machines.

Shift Number

The report can be limited to a single shift number. Only bookings made against this shift in the date range are extracted for inclusion on the report. If the field is left blank, all shifts are included.

Date Range

Enter a date range, or accept the default of the current date.

Report Basis

The option is given to report on labour or machine efficiency.

1 - Machine efficiency. The report compares standard machine hours to actual hours booked to machines.

2 - Labour efficiency. The report compares standard labour hours to actual hours booked to operators or teams. (default)

Include Rework

Indicates whether or not rework activity should be included in the report.

0 - Exclude rework.

1 - Include rework. The report extracts bookings for all activities with a reporting type of 04 (rework held WIP) and 11 (rework operation). This has the effect of increasing the total time taken to produce the same quantity of output, and as a consequence decreases the efficiency of the machine compared with a report excluding rework.



Note: To obtain an analysis of the different reasons for rework, run the Rework Activity Code Analysis option (23/P4P).



Tip: By printing both types of report for the same machines and date range, you are able to quantify the effect of rework efficiency.

Machine Utilisation



This report provides an analysis of how much your production capacity is being used, measured against standard.

The information can help you identify those machines that are not being used to their full potential, representing under-utilisation of the available resources. This may suggest where increased production volumes may be attained.

Non-usage of production resources usually represents overhead costs that are not directly related to the goods normally produced using them. This report provides information enabling you to identify the areas where these overheads occur, and the amount of under-utilisation.

Machine utilisation can be analysed in terms of either labour or machine performance.



Select Machine Utilisation on the menu to display the Machine Utilisation Report Selection screen.

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MJ620                ZS - Style Demonstration Co.           User GBJBAMS      29/03/96
                                                                11:52:35

Machine Utilisation Report Selection

  Organisational Model..? █

  WIP Locations From....?  _  to  _

  Machines From.....?  _  to  _

  Date range from..... 29/03/96 to 29/03/96

  Item.....?  _  (Blank for all)

  Utilisation Basis..... 2  1=Machine, 2=Labour


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F3=Exit  F4=Prompt  ENTER=Submit Job

05-35    S2      MW      KS      IM      II  KENB20A  PET166S2

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Enter a date range, or accept the default of the current date.

Style

If you want the report to compare machine availability with the amount of capacity consumed in the production of a single style, enter an item code in this field. The bookings made against the style in the date range are accumulated and the total printed as a proportion of the total machine capacity.

Utilisation Basis

The option is given to report on labour or machine efficiency.

1 - Machine efficiency. The report compares standard machine hours to actual hours booked to machines.

2 - Labour efficiency. The report compares standard labour hours to actual hours booked to operators or teams. (default)

Setup Cost Report

1... **2...** **3/P4P**

This report details indirect overheads incurred by set up time and shows actual labour and machine time; together with labour, machine and material costs.

- Set up time is recorded when a booking is made against an activity type with a reporting type of:
 - 03 (set up).
- Set up material costs are recorded when material items and quantities are entered in the Materials window during a set up activity booking session.



Select Setup Cost Report on the menu to display the Setup Cost Report Selection screen.

MJ690	ZS - Style Demonstration Co.	User GBJBAMS	29/03/96 11:53:14
Set-up Cost Report Selection			
Organisational Model...? █			
WIP Locations From....? ____ to ____			
Items from.....? _____ to _____			
Date range from..... 29/03/96 to 29/03/96			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35 S2 MW KS IM II KENB20A PET166S2			

Fields:

Organisational Model

Enter the 'live' organisational model.

WIP Locations Range

Enter a range of WIP locations or leave blank to include all locations.

Style Range

Enter a range of styles or leave blank to include all styles.

Date Range

Enter a date range, or accept the default of the current date.

Direct Overhead Variance

1...
2... **4/P4P**

This report compares actual variable overheads incurred for a production quantity at an operation, with standard variable overheads expected for the same quantity. The difference is referred to as the direct overhead variance. Because all overheads are recorded at standard cost, there is no calculation of rate variance; only the total variance is shown.

Direct overheads are a function of the cost centre associated with the machines at which work takes place. The cost centre definition determines whether overhead 1 and/or overhead 2 costs are incurred; and if so, the method used to calculate them. Overhead 1 is always defined as a variable cost; whereas overhead 2 may be variable or fixed.

Where appropriate, (that is, where they have been defined for a machine) direct overheads are calculated and accrued automatically as a consequence of booking work. Indirect overheads may be entered manually through the same routines, by booking against an activity type with an associated reporting type of 06 (indirect overhead); they can be reviewed via the Indirect Overhead Report.



Select Direct Overhead Variance on the menu to display the Direct Overhead Variance Report Selection screen.

MJ675	ZS - Style Demonstration Co.	User GBJBAMS	29/03/96 11:53:49
Direct Overhead Variance Report Selection			
Organisational Model...? __			
WIP Locations from....? __ to __			
Work Stations from....? ____ to ____			
Items from.....? _____ to _____			
Date Range from..... 29/03/96 to 29/03/96			
Include Rework..... 1 1=Yes			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35	SF	MW	KS IM II KENB20A PET166S2

Fields:

Organisational Model

Enter the 'live' organisational model.

WIP Locations Range

Enter a range of WIP locations or leave blank to include all locations.

Machine Range

Enter a range of machines or leave blank to include all machines.

Style Range

Enter a range of styles or leave blank to include all styles.

Date Range

Enter a date range, or accept the default of the current date.

Report Basis

The option is given to report on labour or machine efficiency.

1 - Machine efficiency. The report compares standard machine hours to actual hours booked to machines.

2 - Labour efficiency. The report compares standard labour hours to actual hours booked to operators or teams. (The default).

Include Rework

Rework activity can be included in the report.

0 - Exclude rework.

1 - Include rework. The report extracts bookings for all activities with a reporting type of 04 (rework held WIP) and 11 (rework operation).

This has the effect of increasing the total time taken to produce the same quantity of output, and as a consequence decreases the efficiency of the machine compared with a report excluding rework.



Note: To obtain an analysis of the different reasons for rework, run the Rework Activity Code Analysis option.



Tip: By printing both types of report for the same machines and date range, you are able to quantify the effect of rework efficiency.

Down Time Report

1...
2... **5/P4P**

This report provides an analysis of periods when machines were not operational.

A line is printed for each period of inactive time, giving the date and the number of hours down. This is also expressed as a percentage of the total available time and has a cost associated with it.

Available hours are calculated from the shift profile for the machine and date, and accumulating shift hours; or the default hours if no shift profile has been set.

- Down time is recorded when a booking is made against an activity type with a reporting type of:
 - 05 (down time).



Select Down Time Report on the menu to display the Work Station Downtime Report Selection screen.

MJ695	ZS - Style Demonstration Co.	User GBJBAMS	29/03/96 11:54:24
Work Station Downtime Report Selection			
Organisational Model...? █			
WIP Locations from....? to			
Work Stations from....? to			
Date Range from..... 29/03/96 to 29/03/96			
Reason Code.....? (Blank for all)			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35	82	MW	KS IM II KENB20A PET166S2

Fields:

Organisational Model

Enter the 'live' organisational model.

WIP Locations Range

Enter a range of WIP locations or leave blank to include all locations.

Machine Range

Enter a range of machines or leave blank to include all machines.

Date Range

Enter a date range, or accept the default of the current date.

Reason Code

If the report is to be produced for a single reason code, enter the code here.



Press **Enter** on the Work Station Downtime
Report Selection screen to print the report.

Indirect Overhead Report

1... 6/P4P 2...

This report details indirect overheads incurred at an operation.

- Indirect overheads are recorded when a booking is made against an activity with a reporting type:
 - 06 (indirect overhead).
- Indirect material costs are recorded when material items and quantities are entered in the Materials window during an indirect overhead activity booking session.



Select Indirect Overhead Report on the menu to display the Indirect Overhead Report Selection screen.

MJ685	ZS - Style Demonstration Co.	User GBJBAMS	29/03/96 11:54:51
Indirect Overhead Report Selection			
Organisational Model..? __			
WIP Locations From....? __ to __			
Items from.....? _____ to _____			
Date range from..... <u>29/03/96</u> to <u>29/03/96</u>			
F3=Exit F4=Prompt ENTER=Submit Job			
05-35 S2 MW KS IM II KENB20A PET166S2			

Fields:

Organisational Model

Enter the 'live' organisational model.

WIP Locations Range

Enter a range of WIP locations or leave blank to include all locations.

Style Range

Enter a range of styles or leave blank to include all styles.

Date Range

Enter a date range, or accept the default of the current date.

Direct Labour Variance



11/P4P

Use this report to analyse performance with regard to actual and standard hours booked.

Direct Labour

Direct labour is defined as hours booked to operators/teams against an activity type of:

- 01 (production reporting).
- 04 (rework held WIP).
- 07 (labour booking) at non count point operations.
- 11 (rework at non count point operations).
- 60 (on-std bundle bookings).
- 61 (off-std bundle bookings).
- 62 (own rework bundle bookings).
- 63 (others' rework bundle bookings).

Total Labour Variance

The total labour variance is the difference between the standard labour cost at an operation and the actual labour cost. This is broken down into a rate variance and an efficiency variance. Variances can arise because of differences between the standard and actual rate for the actual number of labour hours booked.

Rate Variance

Rate variance is the difference between the standard labour rate and the actual rate for the actual number of hours booked.

Actual Labour Rate

Actual labour rate is the average rate calculated from the bookings made in the date range.

Labour Efficiency Variance

Labour efficiency variance is the difference between the standard labour hours for the actual production and actual hours, valued at the standard labour rate

Variance Totals

Variance totals are printed for individual operations and are accumulated for each machine. The following information is calculated for each machine:

Value Calculated	Equals
Actual Rate	Actual Cost/Actual Hours
Standard Rate	Std Cost/Std Hours

Value Calculated	Equals
Rate Variance	(Std Rate - Actual Rate) x Actual Hours
Performance Variance	(Std Hours - Actual Hours) x Std Rate
Total Variance	Rate Variance + Performance Variance or Std Cost - Actual Cost



Select Direct Labour Variance on the menu to display the Direct Labour Variance Report screen.

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MJ610          ZS - Style Demonstration Co.      User GBJBAMS      29/03/96
                                                    11:55:18

Direct Labour Variance Report

  Organisational Model...? █
  WIP Locations From....?  to 
  Work Stations From....?  to 
  Items From.....?  to 
  Date From..... 29/03/96 to 29/03/96
  Include Rework..... 1 1=Yes

F3=Exit  F4=Prompt  ENTER=Submit Job

05-35  SP  MW  KS  IM  II  KENB20A  PET166S2
  
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Fields:

Organisational Model

Enter the 'live' organisational model.

WIP Locations Range

Enter a range of WIP locations or leave blank to include all locations.

Machine Range

Enter a range of machines or leave blank to include all machines.

Style Range

Enter a range of styles or leave blank to include all styles.

Date Range

Enter a date range, or accept the default of the current date.

Include Rework

Indicates whether or not rework activity should be included in the report. Enter one of the following:

0 - Exclude rework.

1 - Include rework. This has the effect of increasing the total time taken to produce the same quantity of output, and as a consequence decreases the efficiency of the machine compared with a report excluding rework. The report extracts bookings for all activities with the following reporting types:

- 04 (rework held WIP).
- 11 (rework at non count point operation).
- 62 (own rework bundle bookings).
- 63 (others' rework bundle bookings).



Tip: By printing both types of report for the same machines and date range, you are able to quantify the effect of rework efficiency.



Note: To obtain an analysis of the different reasons for rework, run the Rework Activity Code Analysis option (23/P4P).

Scrap and Rework Analysis



21/P4P

This report analyses the quantities of scrap and rework produced by operations over a range of dates.

- Rework is recorded when a booking is made against an activity with the following reporting types:
 - 04 (rework held WIP).
 - 11 (rework at non count point operations).
 - 62 (own rework bundle bookings).
 - 63 (others' rework bundle bookings).
- Scrap is recorded when a booking is made against an activity with the following reporting types:
 - 01 (production reporting) if a scrap quantity is entered.
 - 34 (scrap at operation) if scrap is recorded at a non count point operation.
 - 32 (scrap held WIP) if WIP inventory booked as held is subsequently scrapped.
 - 33 (scrap WIP) if WIP inventory booked good is then scrapped.
 - 60 (on-std bundle bookings).
 - 61 (off-std bundle bookings).

Rework and scrapped quantities can be recorded against any of the booking routines.



Select Scrap and Rework Analysis on the menu to display the Scrap and Rework Analysis Selection screen.

MJ680	ZS - Style Demonstration Co.	User GBJBMS	29/03/96 11:55:35
Scrap and Rework Analysis Selection			
Organisational Model.....? █			
WIP Locations from.....? __ to __			
Items from.....? _____ to _____			
Date range from..... 29/03/96 to 29/03/96			
F3=Exit F4=Prompt ENTER=Submit Job			
05-41	SE	MW	KS IM II KENB20A PET166S2

Fields:

Organisational Model

Enter the 'live' organisational model.

WIP Locations Range

Enter a range of WIP locations or leave blank to include all locations.

Style Range

Enter a range of styles or leave blank to include all styles.

Date Range

Enter a date range, or accept the default of the current date.

Scrap Reason Code Analysis

1... 22/P4P

This report provides a breakdown of scrapped production by reason code. Whenever scrap is reported, the system requires a reason code to be entered; this may be a default for the entire quantity, or may be specific to a variant quantity.

Totals by reason code are further broken down to give a total scrapped quantity for a reason by item/operation within a user-defined date range.

With this report, you can identify processes that are yielding an abnormally high quantity of scrap material.

Scrap quantities may be entered:

- Directly through standard booking routines.
- By scrapping held WIP.
- By scrapping WIP previously booked good.

Totals are reported for each reason within WIP location.



Select Scrap Reason Code Analysis on the menu to display the Scrap Reason Code Analysis Report Selection screen.

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MJ715          ZS - Style Demonstration Co.      User GBJBAMS      29/03/96
                                                    11:56:04

Scrap Reason Code Analysis Report Selection

Organisational Model.....? █
WIP Locations from.....?  _  to  _
Items from.....?  _  to  _
Date range from..... 29/03/96 to 29/03/96
Reason Code.....?  _

F3=Exit  F4=Prompt  ENTER=Submit Job

05-38  SF  MW  KS  IM  II  KENB20A  PET166S2
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Fields:

Organisational Model

Enter the 'live' organisational model.

WIP Locations Range

Enter a range of WIP locations or leave blank to include all locations.

Style Range

Enter a range of styles or leave blank to include all styles.

Date Range

Enter a date range, or accept the default of the current date.

Reason Code

If the report is to be produced for a single reason code, enter the code here. Only scrap recorded against this code is included on the report. Left blank, the report includes all reason codes.

Rework Activity Code Analysis

1...
2... **23/P4P**

This report provides a breakdown of rework by activity type. Activity types may be defined for different types of rework.

- Rework is recorded when a booking is made against an activity with the following reporting types:
 - 04 (rework held WIP).
 - 11 (rework at non count point operation).
 - 62 (own rework bundle bookings).
 - 63 (others' rework bundle bookings).

These totals are further broken down to give a total rework quantity for an activity by operation. The report enables you to identify production stages that are requiring an abnormally high amount of processing.



Select Rework Activity Code Analysis on the menu to display the Rework Activity Code Analysis Report Selection screen.

MJ720

ST - Style Test & QA Company

User QASTYLE

30/10/97
16:33:10

Rework Activity Code Analysis Report Selection

Organisational Model.....?

WIP Locations from.....? to

Items from.....? to

Date range from..... 30/10/97 to 30/10/97

Activity Code.....?

F3=Exit F4=Prompt ENTER=Submit Job

5-38

SR

MW

KS

CL

IM

DM

II

STUD140

KB

PED201S1

Fields:

Organisational Model

Enter the 'live' organisational model.

WIP Locations Range

Enter a range of locations or leave blank to include all locations.

Style Range

Enter a range of styles or leave blank to include all styles.

Date Range

Enter a date range, or accept the default of the current date.

Activity Code

If the report is to be produced for a single activity code, enter the code here. Only rework recorded against this code is included on the report. Left blank, the report includes all activity codes.